

Chapter 6

The Standard Trade Model

■ Chapter Organization

A Standard Model of a Trading Economy.

Production Possibilities and Relative Supply.

Relative Prices and Demand.

The Welfare Effect of Changes in the Terms of Trade.

Box: U.S. Consumer Gains from Chinese Imports

Determining Relative Prices.

Economic Growth: A Shift of the *RS* Curve.

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Tariffs and Export Subsidies: Simultaneous Shifts in *RS* and *RD*.

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Implications of Terms of Trade Effects: Who Gains and Who Loses?

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Intertemporal Production Possibilities and Trade.

The Real Interest Rate.

Intertemporal Comparative Advantage.

Summary

APPENDIX TO CHAPTER 6: More on Intertemporal Trade

■ Chapter Overview

Previous chapters have highlighted specific sources of comparative advantage that give rise to international trade. This chapter presents a general model that admits previous models as special cases. This “standard trade model” is the workhorse of international trade theory and can be used to address a wide range of issues. Some of these issues, such as the welfare and distributional effects of economic growth, transfers between nations, and tariffs and subsidies on traded goods, are considered in this chapter.

The standard trade model is based upon four relationships. *First*, an economy will produce at the point where the production possibilities curve is tangent to the relative price line (called the isovalue line). *Second*, indifference curves describe the tastes of an economy, and the consumption point for that economy is found at the tangency of the budget line and the highest indifference curve. These two relationships yield the familiar general equilibrium trade diagram for a small economy (one that takes as given the terms of trade), where the consumption point and production point are the tangencies of the isovalue line with the highest indifference curve and the production possibilities frontier, respectively.

You may want to work with this standard diagram to demonstrate a number of basic points. *First*, an autarkic economy must produce what it consumes, which determines the equilibrium price ratio; and *second*, opening an economy to trade shifts the price ratio line and unambiguously increases welfare. *Third*, an improvement in the terms of trade (ratio of export prices to import prices) increases welfare in the economy. *Fourth*, it is straightforward to move from a small country analysis to a two-country analysis by introducing a structure of world relative demand and supply curves, which determine relative prices.

These relationships can be used in conjunction with the *Rybczynski* and the *Stolper-Samuelson theorems* from the previous chapter to address a range of issues. For example, you can consider whether the dramatic economic growth of China has helped or hurt the United States as a whole and also identify the classes of individuals within the United States who have been hurt by China’s particular growth biases. For example, an information box in this chapter provides results from a study that argues that the gains from trade with China are larger for U.S. consumers at the bottom of the income distribution as they tend

to consume relatively more tradable goods (food, apparel, etc.) than richer consumers who spend relatively more on nontraded services. In teaching these points, it might be interesting and useful to relate them to current events. For example, you can lead a class discussion on the implications for the United States of the provision of forms of technical and economic assistance to the emerging economies around the world or the ways in which a world recession can lead to a fall in demand for U.S. exports.

The example provided in the text considers the popular arguments in the media that growth in China hurts the United States. The analysis presented in this chapter demonstrates that the bias of growth is important in determining welfare effects rather than the country in which growth occurs. The existence of biased growth and the possibility of immiserizing growth are discussed. The Relative Supply (*RS*) and Relative Demand (*RD*) curves illustrate the effect of biased growth on the terms of trade. The new terms of trade line can be used with the general equilibrium analysis to find the welfare effects of growth. A general principle that emerges is that a country that experiences export-biased growth will have a deterioration in its terms of trade, while a country that experiences import-biased growth has an improvement in its terms of trade. A case study argues that this is really an empirical question, and the evidence suggests that the rapid growth of countries like China has not led to a significant deterioration of the U.S. terms of trade nor has it drastically improved China's terms of trade. The countries most vulnerable to terms of trade shocks are those that rely on exports of primary commodities with volatile prices. For more diversified economies, the terms of trade tends to be more stable.

An information box in the text discusses the exposure of developing countries to terms of trade shocks and the Covid-19 pandemic. While most countries experience mild swings in their terms of trade some developing nation with export concentrations in mining and agriculture (i.e., commodities) can see substantial changes in their nations welfare given the price volatility of these export goods. One recent study of 38 low-income countries estimated that trade movements accounted for 40 percent of their average GDP fluctuations. The COVID-19 pandemic induced large fluctuations in commodity prices that hit some developing nations very hard. For example, oil prices tanked and even turned negative at one point due to lack of demand. Nigeria, where oil represented 80 percent of all exports, was thrown into the worst recession in four decades.

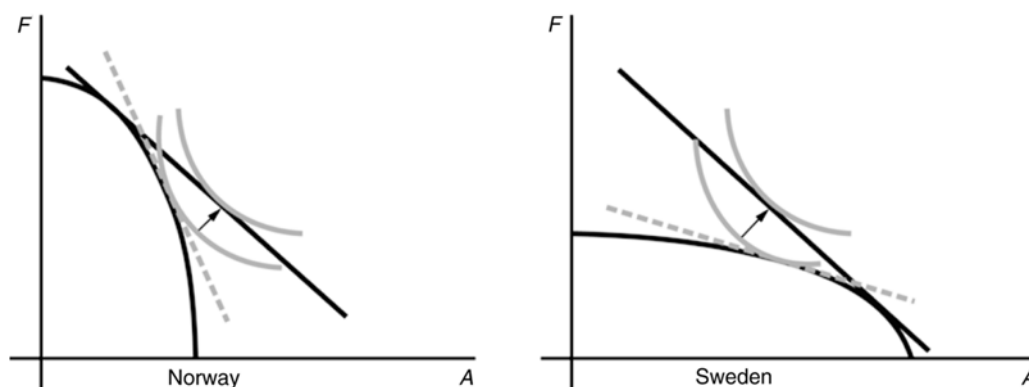
The second area to which the standard trade model is applied is the effects of import tariffs and export subsidies on welfare and terms of trade. The analysis proceeds by recognizing that tariffs or subsidies shift both the relative supply and relative demand curves. A tariff on imports improves the terms of trade, expressed in external prices, while a subsidy on exports worsens terms of trade. The size of the effect depends upon the size of the country in the world. Tariffs and subsidies also impose distortionary costs

upon the economy. Thus, if a country is large enough, there may be an optimum, nonzero tariff. Export subsidies, however, only impose costs upon an economy. Internationally, tariffs aid import-competing sectors and hurt export sectors, while subsidies have the opposite effect.

The chapter then closes with a discussion of international borrowing and lending. The standard trade model is adapted to trade in consumption across time. The relative price of future consumption is defined as $1/(1+r)$, where r is the real interest rate. Countries with relatively high real interest rates (newly industrializing countries with high investment returns, for example) will be biased toward future consumption and will effectively “export” future consumption by borrowing from established developed countries with relatively lower real interest rates.

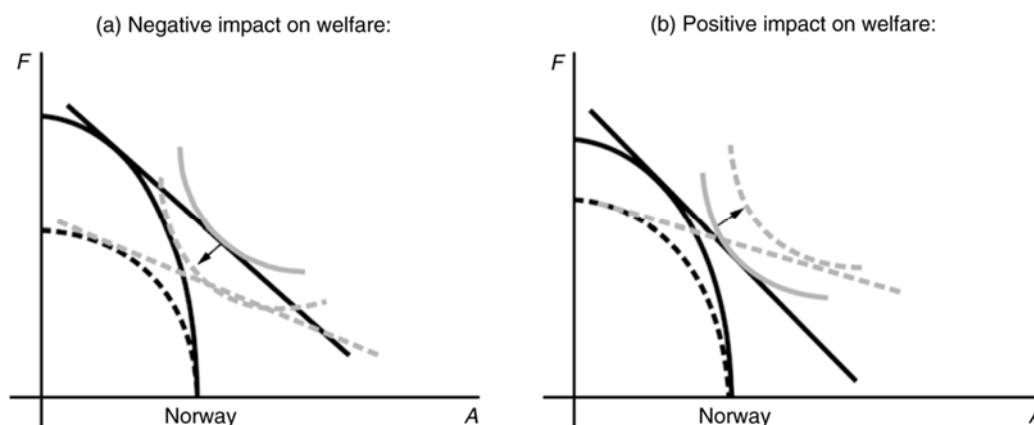
■ Answers to Textbook Problems

1.



Note how welfare in both countries increases as the two countries move from production patterns governed by domestic prices (dashed line) to production patterns governed by world prices (straight line).

2.

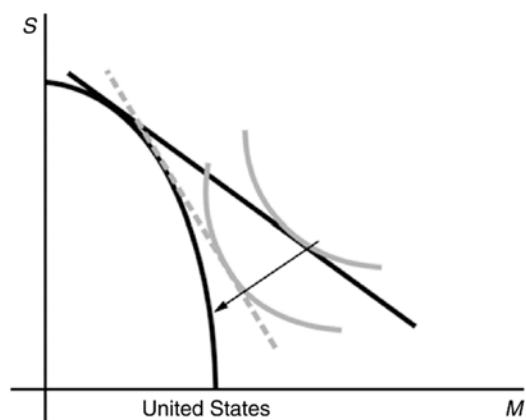


In panel a, the reduction of Norway's production possibilities away from fish cause the production of fish relative to automobiles to fall. Thus, despite the higher relative price of fish exports, Norway moves down to a lower indifference curve representing a drop in welfare.

In panel b, the increase in the relative price of fish shifts causes Norway's relative production of fish to rise (despite the reduction in fish productivity). Thus, the increase in the relative price of fish exports allows Norway to move to a higher indifference curve and higher welfare.

3. An increase in the terms of trade increases welfare when the PPF is right-angled. The production point is the corner of the PPF. The consumption point is the tangency of the relative price line and the highest indifference curve. An improvement in the terms of trade rotates the relative price line about its intercept with the PPF rectangle (because there is no substitution of immobile factors, the production point stays fixed). The economy can then reach a higher indifference curve. Intuitively, although there is no supply response, the economy receives more for the exports it supplies and pays less for the imports it purchases.
4. The difference from the standard diagram is that the indifference curves are right angles rather than smooth curves. Here, a terms of trade increase enables an economy to move to a higher indifference curve. The income expansion path for this economy is a ray from the origin. A terms of trade improvement moves the consumption point further out along the ray.
5. The terms of trade for Japan, a manufactures (M) exporter and a raw materials (R) importer, is the world relative price of manufactures in terms of raw materials (p_M/p_R). The terms of trade change can be determined by the shifts in the world relative supply and demand (manufactures relative to raw materials) curves. Note that in the following answers, world relative supply (RS) and relative demand (RD) are always M relative to R . We consider all countries to be large, such that changes affect the world relative price.

- a. An oil supply disruption from the Middle East decreases the supply of raw materials, which increases the world relative supply of manufactures to raw materials. The world relative supply curve shifts out, decreasing the world relative price of manufactured goods and deteriorating Japan's terms of trade.
 - b. Korea's increased automobile production increases the supply of manufactures, which increases the world RS . The world relative supply curve shifts out, decreasing the world relative price of manufactured goods and deteriorating Japan's terms of trade.
 - c. U.S. development of a substitute for fossil fuel decreases the demand for raw materials. This increases world RD , and the world relative demand curve shifts out, increasing the world relative price of manufactured goods and improving Japan's terms of trade. This occurs even if no fusion reactors are installed in Japan because world demand for raw materials falls.
 - d. A harvest failure in Russia decreases the supply of raw materials, which increases the world RS . The world relative supply curve shifts out. Also, Russia's demand for manufactures decreases, which reduces world demand so that the world relative demand curve shifts in. These forces decrease the world relative price of manufactured goods and deteriorate Japan's terms of trade.
 - e. A reduction in Japan's tariff on raw materials will raise its internal relative price of manufactures (p_M/p_R). This price change will increase Japan's RS and decrease Japan's RD , which increases the world RS and decreases the world RD (i.e., world RS shifts out and world RD shifts in). The world relative price of manufactures declines, and Japan's terms of trade deteriorate.
6. The declining price of services relative to manufactured goods shifts the isovalue line clockwise so that relatively fewer services and more manufactured goods are produced in the United States, thus reducing U.S. welfare.



7. These results acknowledge the biased growth that occurs when there is an increase in one factor of production. An increase in the capital stock of either country favors production of good X, while an increase in the labor supply favors production of good Y. Also, recognize the Heckscher-Ohlin result that an economy will export that good that uses intensively the factor which that economy has in relative abundance. Country A exports good X to country B and imports good Y from country B. The possibility of immiserizing growth makes the welfare effects of a terms of trade improvement due to export-biased growth ambiguous. Import-biased growth unambiguously improves welfare for the growing country.
 - a. The relative price of good X falls, causing country A's terms of trade to worsen. A's welfare may increase or, less likely, decrease, and B's welfare increases.
 - b. The relative price of good Y rises, causing A's terms of trade to improve. A's welfare increases, and B's welfare decreases.
 - c. The relative price of good X falls, causing country B's terms of trade to improve. B's welfare increases, and A's welfare decreases (they earn less for the same quantity of exports).
 - d. The relative price of good X rises, causing country B's terms of trade to worsen. B's welfare may increase or, less likely, decrease, and A's welfare increases.
8. Immiserizing growth occurs when the welfare deteriorating effects of a worsening in an economy's terms of trade swamp the welfare improving effects of growth. For this to occur, an economy must undergo very biased growth, and the economy must be a large enough actor in the world economy such that its actions spill over to adversely alter the terms of trade to a large degree. This combination of events is unlikely to occur in practice.
9. India opening its markets to world trade should be good for the United States if the change reduces the relative price of goods that China sends to the United States and hence increases the relative price of goods that the United States exports. Obviously, any sector in the United States hurt by trade with China would be hurt again by India, but on net, the United States wins. Note that here we are making different assumptions about what India produces and what is tradable than we are in Question 6. Here we are assuming India exports products that the United States currently imports and China currently exports. China will lose by having the relative price of its export good driven down by the increased production in India.
10. What matters for welfare are the external terms of trade. Suppose that country X exports good A and imports good B, while country Y exports good B and imports good A. The export subsidy in country X will raise the internal price of the export good A, leading to an increase in production of good A and decrease demand of good A. As a result, the world price of the good A falls. The tariff on good A

in country Y will increase production and decrease demand for good A in country Y, leading to a reduction in the world price of good A relative to good B. Thus, the terms of trade in country X falls, and the terms of trade in country Y rise. Country X is worse off, while country Y is better off.

If instead country Y had imposed an export subsidy on good B, then the internal price of good B would rise. Production of good B rises, and demand for good B falls. As a result, the world price of good B falls. The net effect on welfare is ambiguous and depends on the relative declines in the world prices of goods A and B.

11. International borrowing and lending implies a trade-off between the production of current and future consumption much like trade in goods implies a trade-off between production of different goods. The more current consumption you select, the less future consumption you will be able to engage in. International borrowing and lending is driven by differences between countries in intertemporal preferences much like international trade is driven by differences between countries in technology or factor endowments. Countries with a relative preference for current consumption will “export” current consumption in exchange for future consumption. In other words, these countries will borrow in the current period from countries that have a relative preference for future consumption. Much like international trade, the relative amount of current consumption that is traded for future consumption is determined by the relative price of future consumption, defined as $1/(1 + r)$, where r is the real interest rate.
12. Comparative advantage in international borrowing and lending is driven by the relative price of future consumption and, more specifically, the real interest rate. As the real interest rate rises, the relative price of future consumption $1/(1 + r)$ falls. Effectively, a country with a high real interest rate is one that has high returns on investment. Such a country will prefer to borrow today and take advantage of the high return on investment and enjoy the fruits of current investment with high returns in the future.
 - a. Countries like Argentina and Canada should have high real interest rates as there are large investment opportunities that have yet to be exploited. These countries will have a low price of future consumption and will be biased toward exporting future consumption, preferring to borrow today.
 - b. Countries like the United Kingdom in the 19th century or the United States today will have relatively lower real interest rates as they already have a high level of capital and limited returns on new investments. As a result, the relative price of future consumption is high, and they will be biased toward exporting present consumption, preferring to lend today.

- c. The discovery of large oil reserves that do not require a significant investment to extract will cause real interest rates in Saudi Arabia to fall (large increase in wealth). This will cause the relative price of future consumption to rise, making it more likely that Saudi Arabia will have a PPF biased toward exporting present consumption. Saudi Arabia will increase their current lending as a result of these oil discoveries.
- d. Oil discovered in Norway that requires a significant investment to extract will have the opposite effect as in answer c. This oil cannot turn into wealth until a significant investment is made, so real interest rates in Norway will rise with this increased demand for investment funds. Higher real interest rates drive the relative price of future consumption down. As a result, Norway will be more likely to export future consumption, borrowing today.
- e. High levels of productivity in South Korea imply that South Korean real interest rates are high given the lucrative investment opportunities in the country. As in answer d, higher real interest rates should drive the relative price of future consumption down and bias South Korea's intertemporal PPF toward exporting future consumption.